

Assignment

Q1. Define Covariance and explain how it differs from Correlation.

Covariance measures how two variables change together.

$$\text{Cov}(X,Y) = \sum (X_i - \bar{X})(Y_i - \bar{Y}) / (n-1)$$

- **Positive covariance:** both variables tend to increase or decrease together.
- **Negative covariance:** when one increases, the other tends to decrease.
- **Zero covariance:** no linear relationship is indicated.

Difference from correlation:

- Covariance has **units and depends on the scale** of the variables.
- Correlation is **unitless** and always lies between **-1 and +1**.
- Correlation is therefore easier to interpret and compare.

Q2. What do positive, negative, and zero covariance indicate?

- **Positive covariance:** XX and YY generally move in the **same direction**.
- **Negative covariance:** XX and YY generally move in **opposite directions**.
- **Zero covariance:** there is **no linear relationship** indicated between the variables.

Q3. Limitations of covariance. Why is correlation preferred?

Covariance has several limitations:

1. Its value depends on the **units and scale** of the variables.
2. A large covariance does not necessarily mean a stronger relationship than a smaller covariance from another dataset.
3. It is difficult to interpret the **strength** of the relationship directly.

Q4. Pearson vs. Spearman correlation

Pearson correlation coefficient:

- Measures the **linear relationship** between two numerical variables.
- Works best when the relationship is approximately linear.
- It can be affected significantly by outliers.

Spearman rank correlation coefficient:

- Measures the strength and direction of a **monotonic relationship** using ranks.
- Does not require the relationship to be linear.
- Is generally less sensitive to outliers.

Prefer Spearman correlation when:

- Data are ordinal/ranked.
 - The relationship is monotonic but not necessarily linear.
 - Data contain outliers or do not satisfy assumptions required for Pearson correlation.
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Q5. Correlation coefficient = 0.85

A correlation of **0.85** indicates a **strong positive linear relationship** between XX and YY.

This means that as XX increases, YY generally tends to increase as well.

However, **we cannot infer causation** from correlation alone. A third variable may influence both XX and YY, or the relationship may be coincidental.

Q6. Calculate covariance between X and Y

Given:

X 2 4 6 8

Y 3 7 5 10

From the dataset:

$$\bar{X} = 2+4+6+8/4 = 5$$

$$\bar{Y} = 3+7+5+10/4 = 6.25$$

Calculate the products of deviations:

$$\sum(X-\bar{X})(Y-\bar{Y})=19$$

Sample covariance:

$$\text{Cov}(X,Y)= 19/(4-1)$$

$$\text{Cov}(X,Y)=6.33$$

Q7. Pearson correlation coefficient between A and B

Given:

A 10 20 30 40 50

B 8 14 18 24 28

Using Pearson's formula:

$$r = \frac{\sum(A - \bar{A})(B - \bar{B})}{\sqrt{\sum(A - \bar{A})^2 \sum(B - \bar{B})^2}}$$

The result is:

$$r \approx 0.998$$

Q8. Correlation between Height and Weight

Given:

Height (cm) 150 160 165 170 180

Weight (kg) 50 55 58 62 70

Using Pearson's correlation formula:

$$r \approx 0.992$$

Q9. Determine whether the correlation is positive or negative

Given:

X 1 2 3 4 5

Y 15 12 9 7 3

As XX increases:

1 → 2 → 3 → 4 → 5

YY decreases:

15 → 12 → 9 → 7 → 3

Q10. Covariance and correlation of the two portfolios

Given:

Year	Portfolio A	Portfolio B
1	8	6
2	10	9
3	12	11
4	9	8
5	11	10

Means:

$$A^- = (8+10+12+9+11)/5 = 10$$

$$B^- = (6+9+11+8+10)/5 = 8.8$$

The sum of the products of deviations is:

$$\sum (A - A^-)(B - B^-) = 12$$

Sample covariance

$$\text{Cov}(A, B) = 12 / (5 - 1)$$

$$\text{Cov}(A, B) = 3$$

The Pearson correlation coefficient is:

$$r \approx 0.986$$